Arista Access Node KR-ACC-24 QoS Congestion Event (Microbursts, Interface Tail Drops, QoS Queue Starvation) Resolution SOP

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6064

Description

This Standard Operating Procedure addresses congestion events, specifically microbursts, interface tail drops, and QoS queue starvation, on the Arista Access Node KR-ACC-24, situated at KR Puram Access. The underlying protocol mechanics driving these issues stem from the node's role as an access point in the network topology. As the node serves as a gateway for subscriber traffic, it is susceptible to congestion due to the bursty nature of internet traffic. This leads to microbursts, where a high volume of packets are transmitted in a short interval, overwhelming the available buffer space in the interface. Consequently, interface tail drops occur, resulting in packet loss and degradation of network performance. Furthermore, QoS queue starvation may arise due to the node's inability to prioritize critical traffic flows, causing delays and jitter. To mitigate these issues, it is essential to employ QoS policies that effectively manage bandwidth allocation, packet scheduling, and policing. By implementing these measures, network administrators can ensure that critical traffic flows are given precedence, minimizing congestion and its associated problems.

Pre-requisite Checks

- Verify that remote hands are dispatched to KR Puram Access for assistance, if required.

- Check the status of redundant links to ensure they are operational and can handle increased traffic loads.

- Confirm that the node's hardware and software configurations are up-to-date and compliant with the latest vendor recommendations.

Potential Root Causes

Cause	Probability
Inadequate QoS configuration or misconfigured traffic policing	High
Insufficient buffer space or inefficient packet processing mechanisms	Medium
Unforeseen changes in network topology or traffic patterns	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware cpu cpu-usage, Expected Output: CPU Usage: 90%, Description: This command isolates CPU usage patterns on the node, helping to identify if high CPU utilization is contributing to congestion.
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```
Command: show interfaces | include Buffer, Expected Output: Buffer: 50%, Description: This command examines the buffer usage on the node's interfaces, allowing administrators to determine if insufficient buffer space is causing congestion.

Command: show qos policy | include policer, Expected Output: Policer: 100%, Description: This command audits the QoS policy configuration on the node, enabling administrators to identify any misconfigured traffic policing or prioritization.
```

Resolution Steps

Step 1: Step: Configure traffic policing on the node's interfaces, Command: configure terminal device-policy police 1000000 1000000, Description: This resolution step sets up traffic policing on the node's interfaces to limit bursty traffic and prevent congestion. By configuring policers, administrators can ensure that critical traffic flows are given precedence, minimizing the risk of packet loss and network performance degradation.

Step 2: Step: Increase buffer space on the node's interfaces, Command: configure terminal device buffer 50000, Description: This resolution step increases the buffer space on the node's interfaces to accommodate higher volumes of traffic. By augmenting buffer capacity, administrators can reduce the likelihood of interface tail drops and packet loss due to congestion.

Step 3: Step: Reconfigure QoS policy to prioritize critical traffic flows, Command: configure terminal device-policy class-map match-any critical-traffic, Description: This resolution step reconfigures the QoS policy to prioritize critical traffic flows, ensuring that they are given precedence over less critical traffic. By implementing this change, administrators can minimize congestion and ensure that the network operates within established service level agreements.

Post-Check Verification

```
Command: show qos policy | include policer, Expected Output: Policer: 100%, Description: This verification command audits the QoS policy configuration on the node to ensure that policers are correctly configured to limit bursty traffic.

Command: show interfaces | include Buffer, Expected Output: Buffer: 50%, Description: This verification command examines buffer usage on the node's interfaces to confirm that buffer space has been increased and is sufficient to handle traffic loads.

Command: show hardware cpu cpu-usage, Expected Output: CPU Usage: 80%, Description: This verification command monitors CPU usage on the node to ensure that it is within acceptable limits and not contributing to congestion.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration, Command: configure terminal device-policy class-map no-match-any critical-traffic, Description: This rollback procedure restores the original QoS policy configuration on the node, reversing any changes made during the resolution process.

- Step: Revert buffer space configuration to original settings, Command: configure terminal device buffer 25000, Description: This rollback procedure reverts the buffer space configuration on the node's interfaces to its original settings, undoing any changes made to increase buffer capacity.

- Step: Revert traffic policing configuration to original settings, Command: configure terminal device-policy police 500000 500000, Description: This rollback procedure reverts the traffic policing configuration on the node's interfaces to its original settings, undoing any changes made to limit bursty traffic.